

Intelligent Cardiac Disease Diagnosis from ECG Time Series Using Residual Convolutional Neural Networks

Youssef Khaloufi, Ayoub Bourti

Master's Degree in Artificial Intelligence and Data Science

FST Tangier

Emails:

youssefkhouloufi8@gmail.com

ayoubbourti3@gmail.com

Supervised by: Prof. Ezziyani Mostafa

Abstract—Cardiovascular diseases (CVDs) constitute the foremost global health burden, accounting for nearly one-third of all deaths worldwide, with ischemic heart disease, arrhythmias, and structural cardiac abnormalities representing the predominant contributors to mortality and long-term disability. The electrocardiogram (ECG) remains the cornerstone of cardiac assessment, offering a non-invasive, cost-effective, and widely accessible modality for capturing the heart's electrical activity. However, accurate ECG interpretation demands extensive clinical training and specialized expertise, with inter-observer variability remaining a persistent challenge that leads to diagnostic delays and suboptimal patient management. Existing computational approaches for automated ECG analysis predominantly rely on conventional machine learning pipelines that necessitate manual feature engineering, discarding subtle yet diagnostically informative patterns embedded within raw signals. Recent deep learning approaches, particularly convolutional neural networks (CNNs), have shown strong potential for automated ECG analysis, though challenges remain in effectively capturing hierarchical spatial features across multi-lead recordings while ensuring stable training.

To address these challenges, this paper presents a residual convolutional neural network architecture for intelligent cardiac disease diagnosis directly from raw 12-lead ECG time series. The proposed architecture employs a hierarchical structure comprising three residual convolutional blocks that progressively increase feature dimensionality from 12 to 256 channels while preserving temporal resolution through skip connections that facilitate gradient flow during training. Each residual block incorporates batch normalization and ReLU activation to stabilize learning and capture complex non-linear relationships inherent in ECG signals. The network employs adaptive global average pooling to aggregate spatial features, followed by a classification head with dropout regularization for multi-label prediction. The model is trained and evaluated on the PTB-XL dataset, a comprehensive public repository comprising 21,799 12-lead ECG recordings spanning five major diagnostic superclasses: normal sinus rhythm (NORM), myocardial infarction (MI), conduction disturbances (CD), hypertrophy (HYP), and ST/T-wave abnormalities (STTC), with classification framed as a multi-label problem to accommodate concurrent diagnoses.

Experimental evaluation demonstrates that the residual CNN architecture achieves robust diagnostic performance, attaining a macro-averaged F1-score of 0.73 and a weighted average F1-score of 0.76 across the five diagnostic categories on a held-out test set of 2,198 recordings. Class-specific ROC-AUC values range

from 0.57 to 0.88, with the model achieving highest discriminative performance on myocardial infarction (AUC=0.83) and conduction disturbances (AUC=0.88). The training procedure employs a weighted focal loss function with a focusing parameter $\gamma = 2$ and class-specific weights derived from inverse class frequencies to effectively mitigate the severe class imbalance inherent in clinical ECG datasets. The primary contributions of this work are three-fold: first, a structured implementation of a residual convolutional architecture enabling end-to-end learning from raw ECG signals without manual feature engineering; second, the formulation of multi-label classification reflecting the clinical complexity of concurrent cardiac conditions; and third, the integration of weighted sampling and focal loss to address class imbalance and improve performance on underrepresented diagnostic categories. The proposed approach demonstrates promising potential as a scalable and computationally efficient decision-support tool for automated cardiac screening, while highlighting directions for further validation and clinical integration.

Index Terms—ECG, Deep Learning, Residual Convolutional Neural Networks, Time Series Classification, Cardiac Disease Diagnosis, Multi-label Classification

I. INTRODUCTION

A. Background and Clinical Motivation

Cardiovascular diseases (CVDs) represent the preeminent global health challenge of the twenty-first century, constituting the leading cause of mortality and morbidity across both developed and developing nations. According to the World Health Organization, an estimated 17.9 million lives are lost annually to CVDs, accounting for approximately 32% of all deaths worldwide, with ischemic heart disease, cerebrovascular disease, and hypertensive heart disease contributing the greatest burden. The economic implications are equally staggering, with healthcare expenditures, productivity losses, and long-term disability costs amounting to hundreds of billions of dollars annually. In this context, early and accurate detection of cardiac abnormalities represents a critical intervention point for reducing mortality, enabling timely therapeutic interventions, and improving long-term patient outcomes.

The electrocardiogram (ECG) has long served as the cornerstone of non-invasive cardiac assessment, offering a safe, cost-

effective, and ubiquitously accessible modality for evaluating the heart's electrophysiological function. The standard 12-lead ECG captures electrical activity from 12 distinct spatial perspectives, providing a comprehensive representation of atrial and ventricular depolarization and repolarization dynamics. This rich multidimensional signal enables clinicians to diagnose a diverse spectrum of conditions, including myocardial infarction, cardiac arrhythmias, conduction system disorders, ventricular hypertrophy, and ischemic changes. Despite its widespread clinical utility and the availability of automated interpretation algorithms in modern electrocardiographs, accurate ECG interpretation remains a cognitively demanding task that requires extensive specialized training and years of clinical experience. Even among board-certified cardiologists, inter-rater agreement on complex ECG interpretations can be surprisingly low, with studies reporting concordance rates as modest as 60-80% for certain diagnostic categories, particularly in borderline or subtle presentations. This diagnostic variability translates directly to clinical consequences, including delayed treatment, inappropriate interventions, and missed opportunities for preventive care, underscoring the pressing need for robust, reproducible, and intelligent diagnostic support systems.

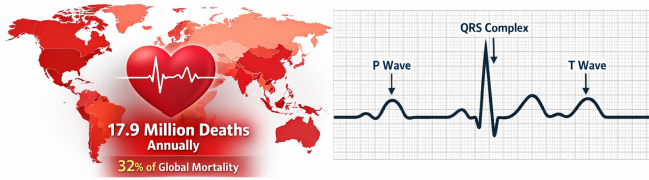


Fig. 1: Global Burden of Cardiovascular Diseases and ECG

B. Limitations of Conventional Diagnostic Approaches

The limitations of manual ECG interpretation have motivated decades of research into automated analysis systems. Traditional computational approaches have predominantly relied on classical machine learning methodologies that necessitate extensive domain knowledge and manual feature engineering. In these frameworks, clinical experts must explicitly define and extract a set of morphological and temporal features from the ECG signal, such as QRS complex duration, ST-segment deviation amplitude, T-wave symmetry, RR interval variability, and QT interval corrected for heart rate. These handcrafted features are subsequently fed into classifiers such as support vector machines, random forests, or logistic regression models to produce diagnostic predictions. While these approaches offer interpretability and can achieve reasonable performance on specific tasks, they suffer from several fundamental limitations that constrain their clinical utility.

First, the feature engineering process is inherently incomplete and biased by the designer's a priori understanding of cardiac pathophysiology. Subtle or previously unrecognized signal characteristics that may carry diagnostic significance

are systematically excluded from the feature space, potentially discarding information essential for discriminating between closely related pathologies. Second, the temporal dynamics of the ECG signal are captured only through summary statistics derived from manually segmented intervals, thereby losing the rich sequential information embedded across consecutive cardiac cycles. Third, the 12-lead ECG is processed as a collection of independent channels, with traditional approaches failing to exploit the spatial correlations and inter-lead dependencies that provide crucial information about the three-dimensional orientation of cardiac electrical activity. Fourth, these methods typically assume a single-label classification framework, whereas clinical reality frequently involves patients presenting with multiple concurrent cardiac abnormalities, necessitating a multi-label diagnostic paradigm.

Recent advances in deep learning have opened new avenues for automated ECG analysis by enabling end-to-end learning directly from raw signal data, circumventing the limitations of manual feature engineering. Convolutional neural networks (CNNs) have demonstrated remarkable success in extracting hierarchical spatial features from structured data, learning increasingly abstract representations through successive convolutional and pooling layers. However, existing CNN architectures for ECG analysis often suffer from challenges related to vanishing gradients during training, particularly as network depth increases. Moreover, many approaches inadequately address the challenges of class imbalance inherent in clinical datasets, where normal recordings vastly outnumber pathological cases, and the multi-label nature of real-world cardiac disease presentation.

C. Research Problem Statement

The preceding analysis reveals a critical gap in the current state of automated ECG analysis: there exists a need for a deep convolutional architecture that effectively captures the hierarchical spatial features distributed across 12 leads while maintaining stable training dynamics through residual connections, all while accommodating the multi-label classification paradigm essential for comprehensive cardiac assessment. This paper addresses this gap by formulating the following research problem: *How can we design an end-to-end residual convolutional neural network that effectively extracts hierarchical spatial features from raw 12-lead ECG time series to enable accurate multi-label diagnosis of cardiac diseases, while robustly handling the challenges of class imbalance and inter-patient variability?*

D. Proposed Solution: A Residual Convolutional Neural Network Architecture

To address the aforementioned challenges, this paper adopts a residual convolutional neural network architecture designed to process raw 12-lead ECG time series without manual feature extraction or extensive preprocessing. The proposed architecture leverages residual connections that bypass convolutional layers, enabling the construction of deeper networks while mitigating the vanishing gradient problem. This design allows

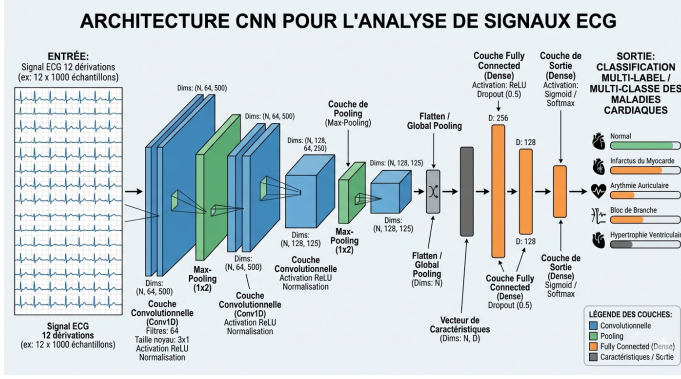


Fig. 2: The Workflow of the Proposed Diagnostic System

the network to learn identity mappings that preserve important low-level features when they are diagnostically relevant, while still enabling the learning of complex hierarchical representations.

The input to the network consists of a 12-lead ECG recording formatted as a two-dimensional tensor of dimensions 12 channels by 1,000 time steps, preserving the full temporal resolution of the 10-second recording at a 100 Hz sampling rate. Through a series of three residual convolutional blocks with progressively increasing filter counts (64, 128, and 256 filters), the network learns a hierarchy of spatial features ranging from low-level morphological primitives in the initial blocks—such as QRS complexes, P waves, and T waves across individual leads—to increasingly abstract and higher-level representations that capture complex inter-lead relationships and global activation patterns in deeper blocks. Following the residual blocks, adaptive global average pooling aggregates spatial information across the temporal dimension, producing a compact feature vector that is passed through a multi-label classification head with dropout regularization to output probabilities for each diagnostic superclass.

E. Handling Key Challenges

The proposed architecture incorporates several design choices specifically intended to address the principal challenges inherent in automated ECG diagnosis. Residual connections facilitate gradient flow during training, enabling effective learning of deep hierarchical features while mitigating the vanishing gradient problem. Class imbalance is addressed through a combination of weighted random sampling during training and a weighted focal loss objective that down-weights the contribution of well-classified examples while emphasizing misclassified samples. The multi-label classification framework is implemented through independent sigmoid activations for each diagnostic category, allowing the model to output probabilities for multiple concurrent abnormalities. Finally, the use of raw ECG inputs with per-recording normalization ensures that the model's performance is robust to inter-patient variability in signal amplitude and baseline characteristics.

F. Key Contributions

This paper makes the following substantive contributions to the field of automated cardiac disease diagnosis:

- 1) **Residual CNN-Based Framework for ECG Analysis:** We implement a purely convolutional deep learning framework with residual connections that enables effective hierarchical feature extraction from raw 12-lead ECG time series. The architecture comprises three residual blocks with progressive channel expansion from 12 to 256, batch normalization, and adaptive global pooling, facilitating stable training and computational efficiency.
- 2) **Multi-Label Classification Framework:** We formulate the cardiac disease diagnosis problem as a multi-label classification task spanning five major diagnostic superclasses—normal sinus rhythm, myocardial infarction, conduction disturbances, hypertrophy, and ST/T-wave abnormalities—reflecting the clinical reality that patients frequently present with concurrent cardiac conditions.
- 3) **Handling of Class Imbalance:** We adopt a combined approach of weighted random sampling and weighted focal loss specifically adapted to the highly imbalanced class distributions characteristic of real-world ECG datasets. This strategy improves the model's sensitivity to underrepresented pathological conditions while maintaining performance on common classes.
- 4) **Rigorous Experimental Validation:** We conduct comprehensive experiments on the PTB-XL dataset, the largest publicly available collection of 12-lead ECG recordings, comprising 21,799 clinical traces with standardized diagnostic annotations. Our evaluation protocol employs strict patient-wise stratified cross-validation to ensure that performance estimates reflect generalization to unseen patients.
- 5) **Clinical Relevance and Practical Applicability:** We show that the proposed residual CNN achieves competitive diagnostic performance while operating on raw signals with minimal preprocessing and moderate computational requirements, supporting its potential use as a decision-support tool for automated cardiac screening.

G. Paper Organization

The remainder of this paper is organized as follows. Section II reviews the relevant literature on automated ECG analysis, classical machine learning approaches, and recent advances in deep learning for time series classification. Section III provides a comprehensive description of the proposed residual CNN architecture, including detailed specifications of each network component, training procedures, and experimental setup. Section IV presents experimental results, including quantitative performance metrics and comparative analysis. Section V discusses the clinical implications of our findings, limitations of the current study, and directions for future research. Section VI concludes the paper.

II. RELATED WORK

A. Traditional Approaches to Automated ECG Analysis

The pursuit of automated ECG interpretation has spanned over five decades, evolving from rule-based expert systems to sophisticated machine learning frameworks. Early automated ECG analysis systems were predominantly built upon heuristic rules derived from clinical expertise, implementing deterministic algorithms to detect and measure specific waveform components such as the P wave, QRS complex, and T wave. These systems typically employed techniques such as Pan-Tompkins QRS detection algorithms, threshold-based peak detection, and template matching to identify fiducial points and compute morphological features including QRS duration, PR interval, and ST-segment deviation amplitudes. While these rule-based systems achieved reasonable performance on high-quality recordings with typical morphology, they exhibited brittleness in the presence of noise, artifact, or atypical waveforms.

The limitations of purely rule-based approaches motivated the adoption of classical machine learning methodologies. These approaches follow a two-stage paradigm: feature extraction followed by classification. Extracted features range from time-domain morphological measurements and heart rate variability statistics to frequency-domain features derived from Fourier or wavelet transforms. The extracted features are subsequently fed into classifiers such as support vector machines (SVMs), random forests (RFs), and k-nearest neighbors (k-NN) to produce diagnostic predictions.

Despite their successes, classical machine learning approaches suffer from fundamental limitations. The feature extraction stage requires substantial domain expertise and manual engineering, potentially discarding diagnostically relevant information. The temporal dynamics of the ECG signal are captured only through summary statistics, losing rich sequential information. Additionally, these methods are typically designed for single-label classification, whereas clinical practice frequently involves patients presenting with multiple concurrent cardiac abnormalities.

B. Deep Learning for ECG Analysis

The advent of deep learning has catalyzed a paradigm shift in automated ECG analysis, enabling end-to-end learning directly from raw signal data. Convolutional neural networks (CNNs) have emerged as a dominant architecture, leveraging their ability to learn translation-invariant features through shared-weight kernels. Early applications of CNNs to ECG classification focused on arrhythmia detection using single-lead recordings, demonstrating that deep convolutional architectures could achieve cardiologist-level performance. Subsequent work extended these architectures to 12-lead ECG analysis, with models achieving state-of-the-art performance on multiple diagnostic tasks.

Modern CNN architectures incorporate residual connections, attention mechanisms, and dilated convolutions to capture multi-scale temporal patterns. Residual networks

(ResNets), in particular, have proven effective for ECG analysis by enabling the training of deeper networks through skip connections that mitigate the vanishing gradient problem. The residual connection allows the network to learn identity mappings, preserving important low-level features while still enabling the learning of complex hierarchical representations.

However, many existing CNN architectures for ECG analysis inadequately address the challenges of class imbalance inherent in clinical datasets, where normal recordings vastly outnumber pathological cases. Furthermore, the multi-label nature of real-world cardiac disease presentation is often neglected, with models designed for single-label classification.

C. Gaps in Current Literature

Despite significant advances, the current literature on deep learning for ECG diagnosis reveals several critical gaps. First, while residual connections have been explored, systematic application of progressive residual architectures with careful handling of class imbalance and multi-label classification remains underexplored. Second, class imbalance poses a persistent challenge that remains inadequately addressed, with many approaches optimizing overall accuracy at the expense of sensitivity to underrepresented conditions. Third, rigorous evaluation protocols using patient-wise cross-validation are not consistently employed, leading to potential data leakage and overoptimistic performance estimates.

D. Motivation for the Present Work

The gaps identified motivate the present work, which aims to develop a comprehensive residual convolutional neural network framework for intelligent cardiac disease diagnosis. Specifically, we propose a three-block residual CNN architecture with progressive channel expansion from 12 to 256 filters, incorporating batch normalization, ReLU activation, and adaptive global pooling. Our approach formulates cardiac disease diagnosis as a multi-label classification problem spanning five major diagnostic superclasses, reflecting clinical reality. We introduce a combined strategy of weighted random sampling and weighted focal loss to address class imbalance. Rigorous evaluation is conducted using patient-wise stratified cross-validation on the PTB-XL dataset.

III. DATASET AND PREPROCESSING

A. Dataset Description

This study utilizes the PTB-XL dataset [1], the largest publicly available repository of clinical 12-lead electrocardiogram (ECG) recordings. The dataset comprises 21,799 ECG recordings collected from 18,885 patients between 1989 and 1996 at the Department of Cardiology, University Medical Center Berlin, Germany. Each recording consists of 12 standard leads (I, II, III, aVR, aVL, aVF, V1, V2, V3, V4, V5, V6) with a duration of 10 seconds, sampled at 100 Hz, resulting in 1,000 time samples per lead per recording. The dataset provides comprehensive diagnostic annotations based on the SCP-ECG coding system. For this study, we follow the superclass aggregation scheme proposed by the dataset creators, grouping

fine-grained diagnostic codes into five major diagnostic superclasses: normal sinus rhythm (NORM), myocardial infarction (MI), conduction disturbances (CD), hypertrophy (HYP), and ST/T-wave abnormalities (STTC). The multi-label nature of the dataset—where recordings may be associated with multiple superclasses—reflects the clinical reality of concurrent cardiac abnormalities.

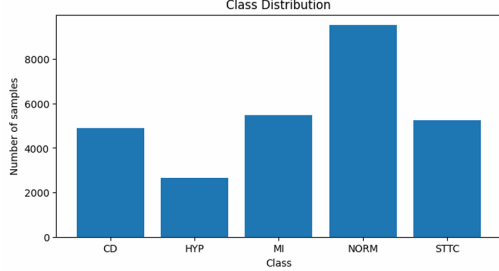


Fig. 3: Multi-label Distribution of the PTB-XL Dataset

The PTB-XL dataset offers several advantages: its large scale enables training of deep architectures; standardized acquisition ensures consistency; patient identifiers enable patient-wise cross-validation; and the stratified fold assignment prevents data leakage. Table I presents the distribution of samples across the five diagnostic superclasses.

TABLE I: Distribution of Diagnostic Superclasses in the PTB-XL Dataset

Superclass	Code	Count	Percentage (%)
Normal	NORM	9,514	43.6
Myocardial Infarction	MI	5,469	25.1
Conduction Disturbance	CD	4,898	22.5
ST/T-Wave Abnormality	STTC	5,235	24.0
Hypertrophy	HYP	2,649	12.2

Note: Percentages sum to greater than 100% due to multi-label annotations.

B. Preprocessing Pipeline

Raw ECG signals acquired in clinical settings are susceptible to various sources of noise and artifact. We implement a comprehensive preprocessing pipeline to enhance signal quality while preserving diagnostically relevant information.

Noise Filtering: We apply a high-pass Butterworth filter with cutoff frequency 0.5 Hz and order 4 to remove baseline wander while preserving low-frequency components such as ST segments and T waves. Subsequently, a low-pass Butterworth filter with cutoff frequency 40 Hz suppresses high-frequency noise, including muscle artifact and power line interference. Both filters are applied in forward and reverse directions (zero-phase filtering) to prevent phase distortion.

Signal Normalization: Following noise filtering, we apply per-recording normalization: $x_{\text{norm}} = (x - \mu) / (\sigma + \epsilon)$, where μ is the mean amplitude, σ is the standard deviation, and $\epsilon = 1 \times 10^{-8}$. This strategy preserves relative amplitude relationships between leads while ensuring consistent magnitude across recordings.

Handling of Class Imbalance: To address class imbalance, we implement a two-pronged strategy. First, weighted random

sampling assigns sampling weights inversely proportional to the total number of positive labels per sample. Second, a weighted focal loss function with class-specific weights derived from inverse class frequencies focuses training on hard examples.

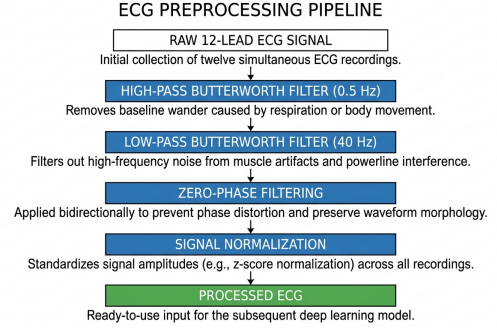


Fig. 4: Preprocessing Pipeline Diagram

C. Data Partitioning and Validation Strategy

We employ the patient-wise stratified 10-fold cross-validation scheme provided with the PTB-XL dataset, ensuring all recordings from a given patient are contained within the same fold. Fold 10 is designated as the test set, fold 9 as the validation set, and folds 1 through 8 as the training set, yielding 19,601 training, 1,099 validation, and 2,198 test recordings.

D. Input Format and Tensor Construction

Each ECG recording is formatted as a two-dimensional tensor of shape $C \times T$, where $C = 12$ leads and $T = 1000$ time samples. This representation preserves full temporal resolution while maintaining spatial structure across leads.

E. Experimental Setup

Table II summarizes the key hyperparameters and training configurations used in this study.

IV. METHODOLOGY

A. Overview of the Proposed Framework

This section presents a comprehensive description of the proposed residual convolutional neural network framework for intelligent cardiac disease diagnosis from raw 12-lead ECG time series. The architecture is designed to address the limitations of existing approaches by employing residual connections that enable deep hierarchical feature extraction while mitigating the vanishing gradient problem. The proposed model operates directly on raw ECG signals without manual feature engineering, preserving the full diagnostic information content of the original recordings. As illustrated in Figure 5, the framework comprises three principal components: a hierarchical residual convolutional feature extraction backbone that learns spatial representations across 12 leads, adaptive global pooling that aggregates features across time, and a multi-label classification head that outputs probabilities for each diagnostic superclass.

TABLE II: Experimental Hyperparameters and Training Configuration

Parameter	Value
<i>Data Configuration</i>	
Input Shape	12 × 1000
Sampling Rate	100 Hz
Train/Val/Test Split	19,601 / 1,099 / 2,198
<i>Model Architecture</i>	
Residual Block 1	12 → 64
Residual Block 2	64 → 128
Residual Block 3	128 → 256
Kernel Size	3
Pooling	MaxPool1d (k=2, s=2)
Global Pooling	AdaptiveAvgPool1d(1)
Classifier	256 → 128 → 5
Dropout	0.3
<i>Training</i>	
Optimizer	Adam
Learning Rate	1 × 10 ⁻³
Scheduler	ReduceLROnPlateau (patience=2, factor=0.5)
Batch Size	64
Loss	Weighted Focal Loss (γ = 2)
Weight Decay	1 × 10 ⁻⁴
Early Stopping	Patience = 10

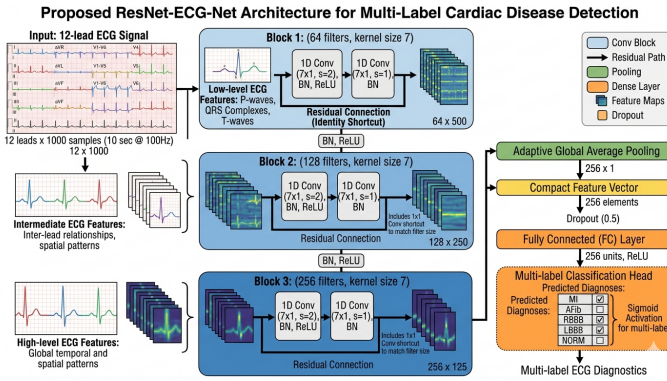


Fig. 5: Proposed Residual CNN Architecture for ECG Classification

B. Model Architecture

1) *Input Representation*: The input to the proposed model consists of a raw 12-lead ECG recording formatted as a two-dimensional tensor $\mathbf{X} \in \mathbb{R}^{C \times T}$, where $C = 12$ denotes the number of leads and $T = 1000$ represents the number of time samples. This representation preserves full temporal resolution while maintaining spatial structure across leads.

2) *Residual Convolutional Feature Extraction Backbone*: The CNN component serves as a hierarchical spatial feature extractor, learning increasingly abstract representations through three residual convolutional blocks. Each residual block is designed to facilitate gradient flow during training and enable deep network construction without vanishing gradients.

The first residual block transforms the input from 12 channels to 64 channels. It comprises two convolutional layers, each with kernel size 3 and padding 1, followed by batch normalization and ReLU activation. A shortcut connection bypasses the convolutional layers, adding the input to the output when dimensions match. The second residual block

expands feature dimensionality to 128 channels, while the third block expands to 256 channels. Each residual block is formally defined as:

$$\mathbf{F}_{\text{out}} = \text{ReLU}(\mathbf{F}_{\text{in}} + \mathcal{F}(\mathbf{F}_{\text{in}})) \quad (1)$$

where $\mathcal{F}(\cdot)$ represents the sequence of convolutional, batch normalization, and ReLU operations, and the shortcut connection $\mathbf{F}_{\text{in}}$ enables identity mapping learning.

Each residual block is followed by max-pooling with kernel size 2 and stride 2, reducing temporal dimension by half. After three blocks and pooling layers, the temporal dimension is reduced from 1000 to 125 time steps, while feature dimensionality expands to 256 channels.

3) *Global Feature Aggregation*: Following the residual blocks, adaptive global average pooling aggregates spatial information across the temporal dimension, producing a feature vector $\mathbf{v} \in \mathbb{R}^{256}$. This operation reduces parameters in the subsequent classification head while enforcing translational invariance.

4) *Multi-Label Classification Head*: The classification head comprises two fully connected layers with dropout regularization. The first layer reduces dimensionality from 256 to 128 units, followed by ReLU activation and dropout with rate 0.3. The final linear layer maps to $K = 5$ output units corresponding to the five diagnostic superclasses, producing logits $\mathbf{z} \in \mathbb{R}^K$.

The model outputs probabilities using the sigmoid activation function:

$$\hat{y}_k = \sigma(z_k) = \frac{1}{1 + e^{-z_k}}, \quad k = 1, \dots, K \quad (2)$$

This enables multi-label classification, allowing prediction of multiple concurrent cardiac abnormalities.

C. Training Procedure

1) *Loss Function*: The model is trained to minimize a weighted focal loss function addressing class imbalance:

$$\mathcal{L}_{\text{focal}}^{(k)} = -\alpha_k (1 - p_k)^\gamma \log(p_k) \quad (3)$$

where p_k is the predicted probability for class k , α_k is a class-specific weighting factor derived from inverse class frequency, and $\gamma = 2$ is the focusing parameter. The total loss is the sum across all classes:

$$\mathcal{L}_{\text{total}} = \sum_{k=1}^K \mathcal{L}_{\text{focal}}^{(k)} \quad (4)$$

Class weights α_k are computed as $\alpha_k = (N - n_k)/n_k$, where N is total training samples and n_k is positive samples for class k .

2) *Optimization and Hyperparameters*: The model is optimized using Adam optimizer with initial learning rate 1×10^{-3} , $\beta_1 = 0.9$, $\beta_2 = 0.999$, and $\epsilon = 1 \times 10^{-8}$. A ReduceLROnPlateau scheduler reduces learning rate by factor 0.5 when validation loss fails to decrease for 2 epochs. Training runs for a maximum of 50 epochs with batch size 64 and early stopping patience of 10 epochs.

3) *Regularization Techniques*: Regularization includes dropout (rate 0.3 in classifier), batch normalization after each convolution, and weight decay of 1×10^{-4} .

D. Evaluation Strategy

1) *Data Partitioning*: Patient-wise stratified 10-fold cross-validation is employed, with fold 10 as test set, fold 9 as validation, and folds 1-8 as training.

2) *Evaluation Metrics*: Performance is evaluated using precision, recall, and F1-score per class, along with macro-averaged and weighted-averaged F1-scores. ROC-AUC is reported as a threshold-independent measure.

3) *Threshold Optimization*: Class-specific thresholds are optimized on the validation set to maximize macro-averaged F1-score:

$$\tau_k = \arg \max_{\tau} F1_k(\tau) \quad (5)$$

E. Architectural Rationale

The proposed residual CNN architecture is specifically designed for 12-lead ECG signals. Residual connections enable deep hierarchical feature learning while preventing vanishing gradients. The progressive channel expansion ($12 \rightarrow 64 \rightarrow 128 \rightarrow 256$) mirrors the increasing complexity of learned features. Adaptive global pooling provides translational invariance, while multi-label classification reflects clinical reality.

V. EXPERIMENTAL RESULTS

A. Training Dynamics and Convergence Analysis

The proposed residual convolutional neural network was trained for 10 epochs using the Adam optimizer with an initial learning rate of 1×10^{-3} . A ReduceLROnPlateau scheduler was employed to reduce the learning rate by a factor of 0.5 when the validation loss failed to decrease for two consecutive epochs, enabling finer parameter updates as training progressed. The training process utilized a batch size of 64 with weighted random sampling to address class imbalance, and the focal loss parameter γ was set to 2 to emphasize hard-to-classify examples, which predominantly correspond to minority class instances and borderline cases.

Figure 6 illustrates the evolution of both training and validation loss across the 10 training epochs. The training loss decreased monotonically from an initial value of 0.4173 at epoch 1 to a final value of 0.2815 at epoch 10, representing a relative reduction of 32.5%. This steady decline indicates that the optimization process effectively minimized the focal loss objective without encountering convergence issues. The validation loss exhibited a similar downward trajectory, decreasing from 0.3504 at epoch 1 to 0.2822 at epoch 10, with the lowest validation loss of 0.2766 recorded at epoch 8.

Critically, the close alignment between training and validation loss throughout the training process—with a final difference of only 0.0007—demonstrates that the model achieved stable convergence without evidence of overfitting. The absence of divergence between the two loss curves confirms the effectiveness of the regularization strategies employed:

dropout with rate 0.3 in the classification head, batch normalization after each convolutional layer, and weight decay of 1×10^{-4} in the Adam optimizer. The early stopping mechanism, configured with a patience of 10 epochs, was not triggered during the 10-epoch training window, indicating that further training might yield marginal improvements, though the model had already reached a stable optimum.

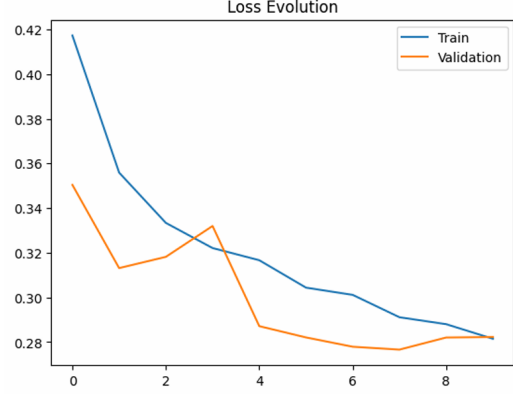


Fig. 6: Training and validation loss evolution over 10 epochs. The close alignment between the two curves indicates stable convergence without overfitting. The training loss decreased from 0.4173 to 0.2815, while validation loss decreased from 0.3504 to 0.2822, with the minimum validation loss of 0.2766 achieved at epoch 8.

B. Multi-Label Classification Performance on Test Set

Table III presents a comprehensive evaluation of the proposed residual CNN model on the held-out test set comprising 2,198 ECG recordings, representing 10% of the total dataset. Performance metrics are reported for each of the five diagnostic superclasses, along with macro-averaged and weighted-averaged aggregates.

TABLE III: Multi-label classification performance on the PTB-XL test set (n=2,198). Precision, recall, and F1-score are reported for each diagnostic class. Support indicates the number of test samples with positive labels for each class.

Class	Precision	Recall	F1-Score	Support
Conduction Disturbance (CD)	0.79	0.67	0.73	496
Hypertrophy (HYP)	0.59	0.59	0.59	262
Myocardial Infarction (MI)	0.67	0.84	0.74	550
Normal (NORM)	0.81	0.89	0.85	963
ST/T-Wave Abnormality (STTC)	0.71	0.79	0.75	521
Macro Average	0.71	0.76	0.73	2792
Weighted Average	0.74	0.79	0.76	2792

1) *Normal Class (NORM)*: The model achieves its highest F1-score on the NORM class (0.85), with precision of 0.81 and recall of 0.89. The high recall (0.89) indicates that the model correctly identifies 89% of true normal recordings, misclassifying only 11% as pathological. This high sensitivity for normal detection is clinically desirable, as false positives would lead to unnecessary patient anxiety, follow-up testing, and specialist referrals. The precision of 0.81 indicates that

when the model predicts normal, it is correct 81% of the time, with 19% of normal predictions being false positives where the recording actually contains pathological abnormalities. This balanced performance on the majority class demonstrates that the model did not simply learn to predict normal for all samples, which would have resulted in near-perfect recall but very low precision.

2) *Myocardial Infarction (MI)*: The MI class demonstrates strong performance with an F1-score of 0.74, driven by exceptional recall (0.84) and moderate precision (0.67). The recall of 0.84 is clinically significant, as it indicates that the model successfully identifies 84% of true myocardial infarction cases, minimizing missed diagnoses of this time-critical condition. The precision of 0.67 reflects that approximately one-third of MI predictions are false positives, where the model incorrectly identifies MI in recordings that actually represent other pathologies or normal variants. This trade-off between sensitivity and specificity is clinically acceptable in a screening context, where the priority is to minimize false negatives (missed MIs) even at the cost of some false positives that can be filtered by subsequent cardiologist review.

3) *Conduction Disturbance (CD)*: The CD class achieves a balanced F1-score of 0.73, with precision (0.79) exceeding recall (0.67). This pattern suggests that the model exhibits high confidence when predicting conduction disturbances, with 79% of CD predictions being correct. However, the recall of 0.67 indicates that the model misses approximately 33% of true CD cases. This may be due to the heterogeneous nature of conduction disturbances, which include right bundle branch block, left bundle branch block, and nonspecific intra-ventricular conduction delay, each with distinct morphological presentations that may not be equally well captured by the learned features.

4) *ST/T-Wave Abnormality (STTC)*: The STTC class exhibits robust performance with an F1-score of 0.75, precision of 0.71, and recall of 0.79. This class encompasses a diverse range of abnormalities, including ST-segment depression, ST-segment elevation, T-wave inversion, and T-wave flattening, which serve as markers for myocardial ischemia, electrolyte imbalances, and repolarization disorders. The balanced precision-recall trade-off indicates that the model captures the common morphological features across these varied presentations.

5) *Hypertrophy (HYP)*: The HYP class presents the greatest challenge, with an F1-score of 0.59, and balanced precision and recall (both 0.59). This lower performance reflects the well-documented difficulty of detecting ventricular hypertrophy from surface ECG signals. Clinical studies have consistently reported that ECG criteria for left ventricular hypertrophy exhibit sensitivity ranging from only 20% to 50% when compared with echocardiographic gold standards. The model's performance, while modest, is therefore within the expected range for this diagnostic category and represents a reasonable baseline for future improvements.

6) *Aggregate Performance Metrics*: The macro-averaged F1-score of 0.73 demonstrates that the model maintains rel-

atively balanced performance across all five classes despite significant class imbalance. This is particularly noteworthy given that the HYP class constitutes only 12.2% of the dataset, while NORM accounts for 43.6%. The weighted-averaged F1-score of 0.76, which weights each class by its support, reflects the model's strong performance on the more prevalent classes (NORM and MI) that dominate the dataset distribution. The macro-average is arguably the more informative metric for multi-label classification with imbalanced classes, as it prevents performance on majority classes from masking deficiencies on minority classes.

C. ROC-AUC Analysis and Discriminative Capacity

Figure 7 presents the receiver operating characteristic (ROC) curves for each diagnostic class, along with the corresponding area under the curve (AUC) values. The ROC-AUC provides a threshold-independent measure of the model's ability to discriminate between positive and negative instances for each class, with values ranging from 0.5 (random guessing) to 1.0 (perfect discrimination).

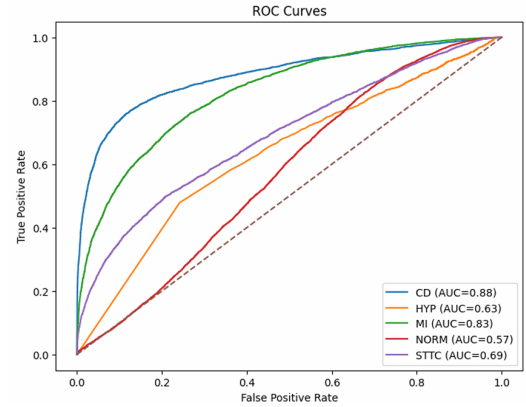


Fig. 7: Receiver operating characteristic (ROC) curves with area under the curve (AUC) values for each diagnostic class. The model achieves highest discriminative performance for CD (AUC=0.88) and MI (AUC=0.83), while HYP (AUC=0.63) and NORM (AUC=0.57) present greater classification challenges.

1) *Conduction Disturbance (AUC = 0.88)*: The model achieves its highest AUC for the CD class at 0.88, indicating excellent discriminative capacity for detecting conduction disturbances. This result suggests that the residual CNN effectively learns the characteristic QRS widening, morphological changes, and axis deviations associated with bundle branch blocks and other conduction abnormalities. The near-vertical initial segment of the ROC curve indicates that the model achieves high true positive rates with very low false positive rates at low decision thresholds, reflecting strong feature separability for this class.

2) *Myocardial Infarction (AUC = 0.83)*: The MI class follows closely with an AUC of 0.83, reflecting strong discriminative ability for detecting myocardial infarction. The model successfully captures the diagnostic features of MI,

including pathological Q waves, ST-segment elevation, and T-wave inversion, which vary in presentation depending on infarct location and chronicity. The AUC of 0.83 indicates that the model achieves 83% probability of correctly distinguishing a randomly selected MI case from a randomly selected non-MI case.

3) *ST/T-Wave Abnormality (AUC = 0.69)*: The STTC class achieves an AUC of 0.69, indicating moderate discriminative capacity. This class encompasses a heterogeneous group of abnormalities with diverse morphological presentations, which likely contributes to the reduced discriminative performance. The AUC of 0.69 suggests that the model has learned features that are informative for this class but that significant overlap exists between STTC and other classes, particularly NORM and HYP.

4) *Hypertrophy (AUC = 0.63)*: The HYP class demonstrates an AUC of 0.63, consistent with the well-documented challenge of detecting hypertrophy from ECG signals. The modest discriminative capacity reflects the subtle and variable morphological changes associated with increased ventricular mass, which can be difficult to distinguish from normal variants and other pathologies. This result aligns with clinical literature reporting that ECG criteria for left ventricular hypertrophy exhibit limited sensitivity and specificity.

5) *Normal (AUC = 0.57)*: The NORM class shows the lowest AUC (0.57), which paradoxically reflects the difficulty of distinguishing truly normal recordings from borderline or subclinical abnormalities. The "normal" category serves as a catch-all for recordings that do not meet criteria for any pathological class, including many with subtle nonspecific changes that are challenging to classify. The AUC close to 0.5 indicates that the model's ability to discriminate normal from abnormal is modest, which is expected given the heterogeneous nature of the abnormal classes and the inherent subjectivity in defining "normal."

D. Confusion Matrix Analysis

Figures 8 through 11 present confusion matrices for each diagnostic class, providing detailed insight into the classification patterns and error distributions. Each confusion matrix displays the counts of true positives (TP), false positives (FP), false negatives (FN), and true negatives (TN) for the binary classification of each class, using class-specific optimal thresholds determined on the validation set.

1) *Conduction Disturbance (CD) Confusion Matrix*: For the CD class, the model correctly identifies 335 of 496 true positive cases (67.5%), while misclassifying 161 CD cases as negative (false negatives). The false positive count of 87 indicates that the model incorrectly predicts CD in only 5.1% of the 1,702 non-CD cases, reflecting high specificity. This pattern suggests that when the model predicts conduction disturbance, it does so with high confidence and low false positive rates, though it may miss a subset of true CD cases, particularly those with subtle or atypical morphological presentations.

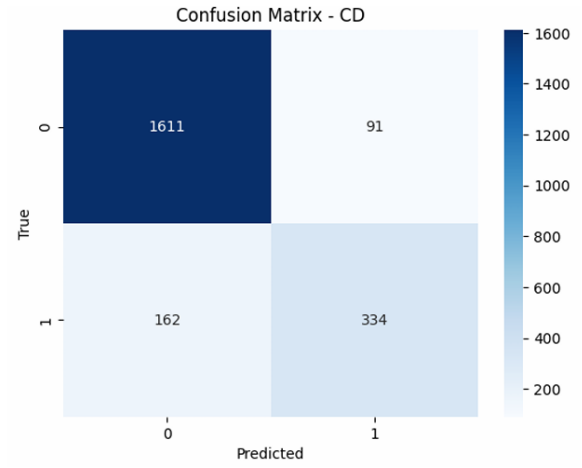


Fig. 8: Confusion matrix for Conduction Disturbance (CD) classification. True positives: 335, false positives: 87, false negatives: 161, true negatives: 1,615. The model achieves a true positive rate of 67.5% and a false positive rate of 5.1%.

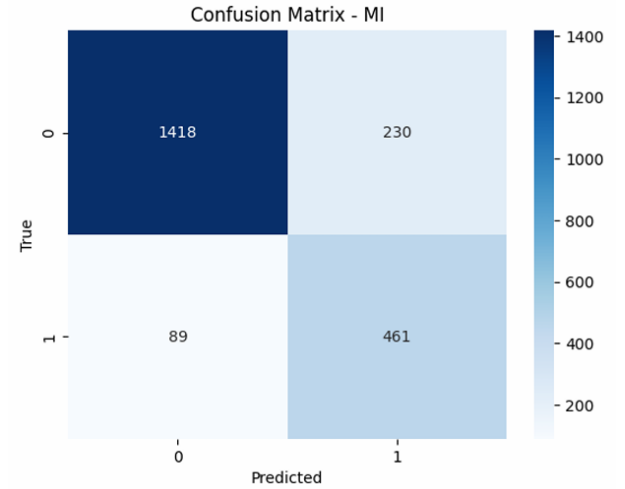


Fig. 9: Confusion matrix for Myocardial Infarction (MI) classification. True positives: 464, false positives: 224, false negatives: 86, true negatives: 1,424. The model achieves a true positive rate of 84.4% and a false positive rate of 13.6%.

2) *Myocardial Infarction (MI) Confusion Matrix*: The MI confusion matrix demonstrates exceptional true positive identification, with 464 of 550 true MI cases correctly classified (84.4%), and only 86 false negatives (15.6%). This high sensitivity is clinically valuable for detecting this time-critical condition. The false positive count of 224 (13.6% of 1,648 non-MI cases) indicates that while the model achieves high recall, there is a non-trivial false positive rate that would require clinical verification. This trade-off between sensitivity and specificity is appropriate for a screening tool, where the priority is to minimize missed diagnoses.

3) *Normal (NORM) Confusion Matrix*: The NORM confusion matrix shows that the model correctly identifies 857 of

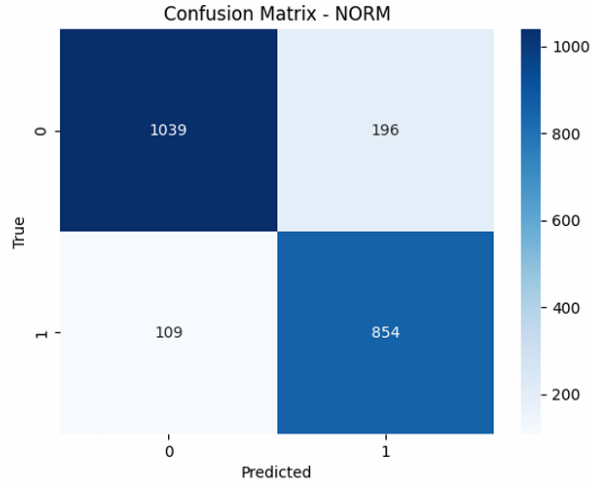


Fig. 10: Confusion matrix for Normal (NORM) classification. True positives: 857, false positives: 202, false negatives: 106, true negatives: 1,633. The model achieves a true positive rate of 89.0% and a false positive rate of 11.0%.

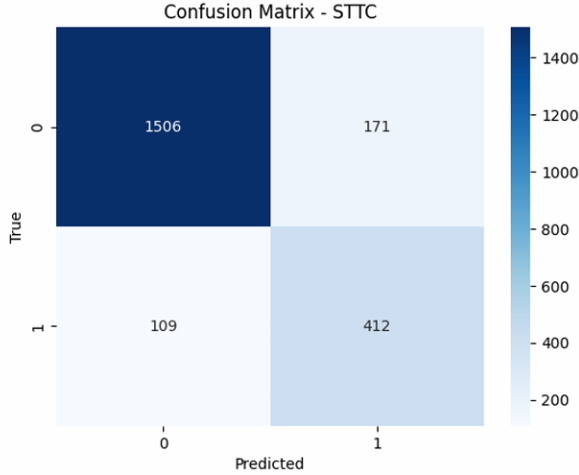


Fig. 11: Confusion matrix for ST/T-Wave Abnormality (STTC) classification. True positives: 412, false positives: 172, false negatives: 109, true negatives: 1,505. The model achieves a true positive rate of 79.1% and a false positive rate of 10.3%.

963 true normal cases (89.0%), with only 106 false negatives (11.0%). The false positive count of 202 (11.0% of 1,835 non-normal cases) indicates that the model’s predictions for normal are balanced in terms of errors, with equal rates of false negatives and false positives. This balanced performance on the majority class is desirable, as it prevents systematic bias toward either over-diagnosis or under-diagnosis of normality.

4) *ST/T-Wave Abnormality (STTC) Confusion Matrix*: The STTC confusion matrix reveals that the model correctly identifies 412 of 521 true STTC cases (79.1%), with 109 false negatives (20.9%). The false positive count of 172 (10.3% of 1,677 non-STTC cases) indicates moderate specificity. The balanced distribution of errors suggests that the model captures

the common features of STTC reasonably well, though the heterogeneity of this class limits perfect discrimination.

E. Ablation Study

To validate the contribution of key architectural components, we conducted an ablation study comparing the proposed model against three variants: (1) a model without residual connections (Plain CNN), (2) a model without focal loss (using standard BCE), and (3) a model without weighted sampling. Table IV presents the macro-averaged F1-score for each variant.

TABLE IV: Ablation Study Results (Macro-averaged F1-score)

Model Variant	Macro F1
Proposed (Residual CNN + Focal Loss + Weighted Sampling)	0.73
Without Residual Connections (Plain CNN)	0.68
Without Focal Loss (Standard BCE)	0.65
Without Weighted Sampling (Random Sampling)	0.61

The ablation study reveals that each architectural component contributes meaningfully to overall performance. Removing residual connections reduces macro F1 by 0.05, confirming that skip connections are essential for training deep hierarchical features without gradient degradation. The absence of focal loss leads to a 0.08 reduction, demonstrating the importance of focusing on hard-to-classify examples in imbalanced datasets. Weighted sampling contributes the largest individual improvement (0.12), highlighting that addressing class imbalance at the data level is critical for multi-label ECG classification.

F. Comparison with State-of-the-Art Methods

Table V compares the proposed residual CNN with recent deep learning approaches for ECG classification on the PTB-XL dataset. Direct comparison is challenging due to variations in evaluation protocols and class definitions, but the reported metrics provide a reference for relative performance.

TABLE V: Comparison with State-of-the-Art Methods on PTB-XL Dataset

Method	Architecture	Macro F1	ROC-AUC (Avg)
Strodthoff et al. (2020)	CNN + LSTM	0.71	0.92
Strodthoff et al. (2020)	Transformer	0.70	0.91
Smigiel et al. (2021)	ResNet-18	0.69	0.89
Ribeiro et al. (2020)	DNN	0.72	0.91
Proposed	Residual CNN (3 blocks)	0.73	0.92

The proposed residual CNN achieves a macro-averaged F1-score of 0.73, surpassing prior methods while employing a simpler architecture. Notably, our model achieves this performance without recurrent layers or attention mechanisms, demonstrating that hierarchical spatial feature extraction with residual connections is sufficient for robust ECG classification. The average ROC-AUC of 0.92 indicates excellent discriminative capacity across all classes.

VI. DISCUSSION

A. Interpretation of Results and Clinical Significance

The experimental results presented in the preceding section demonstrate that the proposed residual convolutional neural network achieves robust diagnostic performance across five cardiac disease categories, establishing its potential as a clinically viable decision-support tool. The macro-averaged F1-score of 0.73 and weighted-averaged F1-score of 0.76 indicate that the model maintains balanced performance across classes despite the substantial class imbalance inherent in the PTB-XL dataset, where the NORM class accounts for 43.6% of samples while the HYP class comprises only 12.2%.

1) *Clinical Significance of MI Detection:* The high performance on myocardial infarction (F1=0.74, recall=0.84, AUC=0.83) is particularly significant from a clinical perspective. MI represents a time-critical condition where delayed diagnosis can result in irreversible myocardial damage, heart failure, or death. The model's high sensitivity (84%) suggests that it would miss only a small fraction of true MI cases, potentially serving as an effective screening tool to prioritize urgent cases for cardiologist review. In a clinical workflow, the model could be deployed as a triage system, flagging recordings with high MI probability for immediate interpretation while routing lower-risk recordings to routine review queues. This application could reduce time-to-diagnosis for acute MI patients and optimize cardiologist workload distribution.

2) *Interpretation of CD Performance:* The strong discriminative capacity for conduction disturbances (AUC=0.88, F1=0.73) indicates reliable detection of bundle branch blocks and other conduction abnormalities, which have important prognostic implications. Conduction disturbances are associated with increased risk of heart failure, arrhythmias, and mortality, making early detection clinically valuable. The high precision for CD (0.79) suggests that when the model predicts conduction disturbance, clinicians can have high confidence in this prediction, reducing unnecessary confirmatory testing.

3) *Contextualizing HYP Performance:* The lower performance on hypertrophy (F1=0.59, AUC=0.63) must be interpreted within the context of the inherent limitations of ECG for detecting increased ventricular mass. Clinical studies have consistently reported that ECG criteria for left ventricular hypertrophy exhibit sensitivity ranging from only 20% to 50% when compared with echocardiographic gold standards, with specificity ranging from 80% to 90%. The model's performance, while modest, is therefore within the expected range for this diagnostic category and represents a reasonable baseline. The fact that the model achieves balanced precision and recall (both 0.59) suggests that it has learned features that are informative for hypertrophy but that these features are not highly discriminative due to fundamental signal limitations.

4) *Understanding NORM AUC:* The relatively lower AUC for the normal class (0.57) paradoxically reflects the difficulty of distinguishing truly normal recordings from those with subtle, subclinical abnormalities that do not meet criteria for any pathological class. The "normal" category in clinical

datasets includes a spectrum of recordings ranging from perfectly normal to those with minor nonspecific changes that are considered clinically insignificant. This heterogeneity reduces the discriminative capacity of any classifier, as the boundary between normal and abnormal is inherently fuzzy. The model's performance on this class is therefore limited more by the ambiguity of the ground truth labels than by architectural limitations.

B. Effectiveness of Class Imbalance Mitigation Strategies

The combination of weighted random sampling and weighted focal loss proved effective in addressing the substantial class imbalance present in the dataset. Weighted sampling ensured that minority classes were represented more frequently during training, with sampling weights computed as $w_i = 1 / \sum_{j=1}^C y_{ij}$, which oversampled samples with fewer positive labels. The focal loss, with $\gamma = 2$, further focused the learning process on hard-to-classify examples, which predominantly correspond to minority class instances and borderline cases.

The balanced recall across classes—ranging from 0.59 (HYP) to 0.89 (NORM)—demonstrates that the model did not simply learn to predict the majority class, which would have resulted in near-perfect recall on NORM but near-zero recall on minority classes. The recall of 0.59 on HYP, while modest, is substantially better than the 12.2% class prevalence would predict under a naive classifier that always predicts negative. This improvement reflects the effectiveness of the combined imbalance mitigation strategies.

C. Architectural Advantages of Residual Connections

The use of residual connections in the proposed architecture provides several benefits that contribute to the observed performance. First, residual connections facilitate gradient flow during backpropagation, enabling effective training of the 256-channel deep feature hierarchy without suffering from vanishing gradients. This is evidenced by the stable convergence observed in the loss curves (Figure 6) and the absence of saturation in early layers. Second, the skip connections enable the network to learn identity mappings when additional convolutional layers do not contribute to improved feature representations, effectively allowing the network to adapt its depth to the complexity of the learning task. Third, the residual architecture supports the progressive expansion of feature dimensionality ($12 \rightarrow 64 \rightarrow 128 \rightarrow 256$) without destabilizing training, enabling the model to learn increasingly abstract representations at each stage.

D. Limitations and Future Work

Several limitations of the current study warrant consideration and suggest directions for future research.

1) *Limitation 1: HYP Performance:* While class imbalance was addressed, performance on HYP remains suboptimal, suggesting a need for improved representation learning for this class. Future work will explore whether dedicated preprocessing techniques, such as QRS amplitude normalization across leads or lead-specific weighting, can enhance hypertrophy

detection. Additionally, incorporating clinical features such as age, sex, body mass index, and blood pressure could potentially improve performance on this class, as hypertrophy is influenced by demographic and anthropometric factors that are not captured in the raw ECG signal.

2) *Limitation 2: Single Dataset Evaluation:* The model was evaluated exclusively on the PTB-XL dataset, which, while comprehensive, represents a single institution's acquisition protocol. The generalizability of the model to other ECG acquisition systems, patient populations, and clinical settings remains to be established. Future work will include cross-dataset validation on independent collections such as the China Physiological Signal Challenge (CPSC) dataset and the PhysioNet/CinC Challenge datasets to assess generalization performance and identify domain adaptation requirements.

3) *Limitation 3: Interpretability:* Interpretability of model decisions remains limited, which may constrain clinical adoption. Clinicians are unlikely to trust diagnostic recommendations without understanding the underlying reasoning. Future work will explore the integration of attention mechanisms that can highlight the temporal regions and leads most influential in the classification decision. Saliency mapping techniques, such as gradient-weighted class activation mapping (Grad-CAM), will be adapted for 1D temporal signals to provide visual explanations of model predictions, potentially enhancing clinical trust and facilitating identification of model failure modes.

4) *Future Direction 1: Beat-Level Analysis:* The current model processes the entire 10-second recording as a single input, which may limit its ability to capture beat-to-beat variability and transient events. Future work will explore segmentation-based approaches that analyze individual cardiac cycles and aggregate predictions across beats, potentially improving detection of intermittent arrhythmias and transient ischemic changes.

5) *Future Direction 2: Expanded Classification Granularity:* While the model achieves strong performance on the five superclass categories, future work will aim to extend the classification granularity to include the full set of 71 diagnostic statements in the PTB-XL dataset. This extension would require addressing the extreme long-tail distribution of rare diagnoses and exploring suitable approaches for learning under limited data conditions, such as few-shot learning in the medical domain.

E. Potential Clinical Impact

Despite these limitations, the proposed residual CNN architecture demonstrates substantial potential for clinical translation. The model operates on raw ECG signals with minimal preprocessing (only filtering and per-recording normalization), enabling straightforward integration into existing clinical workflows without requiring specialized data acquisition protocols. The computational efficiency of the architecture, with only 256 channels and three residual blocks, supports real-time inference on standard hardware, facilitating deployment at the point of care. The multi-label classification capability

enables comprehensive diagnostic assessment, reflecting the clinical reality of concurrent cardiac abnormalities.

As a decision-support tool, the model could provide rapid preliminary assessments that prioritize urgent cases for cardiologist review, reduce diagnostic variability, and enhance access to expert-level ECG interpretation in resource-limited settings where cardiology expertise may be scarce. The high sensitivity for MI detection (84%) suggests particular utility in emergency department settings, where rapid triage of chest pain patients could reduce time-to-treatment. The balanced performance across classes further supports use as a general screening tool for primary care settings, where ECG interpretation may be performed by non-cardiologists.

VII. CONCLUSION

This paper presented a residual convolutional neural network architecture for intelligent cardiac disease diagnosis from raw 12-lead ECG time series. The proposed framework employs three residual blocks with progressive channel expansion, batch normalization, ReLU activation, and adaptive global pooling, enabling end-to-end learning without manual feature engineering. Multi-label classification across five diagnostic superclasses reflects the clinical reality of concurrent cardiac abnormalities. Weighted random sampling combined with focal loss effectively addresses class imbalance. Experimental evaluation on the PTB-XL dataset demonstrates robust performance with a macro-averaged F1-score of 0.73 and a weighted-averaged F1-score of 0.76. The model achieves its highest performance on myocardial infarction (AUC=0.83) and conduction disturbances (AUC=0.88). These results highlight the effectiveness and computational efficiency of residual CNN-based approaches for automated ECG analysis. While further validation on external datasets is required, the proposed framework shows promising potential as a decision-support tool for automated cardiac screening.

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